$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\boxed{\det(A) = (a \times d) - (b \times c)}$$

$$A = \begin{bmatrix} 3 & 4 \\ 1 & 5 \end{bmatrix}$$

$$\det(A) = 3 \times 5 - 4 \times 1$$

$$= 15 - 4$$

$$= 11$$

$$B = \begin{bmatrix} -2 & 7 \\ 3 & 0 \end{bmatrix}$$

$$\det(B) = -2 \times 0 - 3 \times 7$$

$$= 0 - 21$$

$$= -21$$

$$M = \begin{bmatrix} x+1 & 2x \\ 3 & x-2 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a: x+1 & b: 2x \\ c = 3 & d: x-2 \end{bmatrix}$$

$$\det(M) = (x+1)(x-2) - (2x)(3)$$

$$= x^2 - 2x + 1x - 2 - 6x$$

$$= x^2 - 1x - 2 - 6x$$

$$= \boxed{x^2 - 7x - 2}$$

Exercício:

Dada a matriz

$$A = \begin{bmatrix} x & 3 \\ 2 & x+1 \end{bmatrix},$$

determine todos os valores de x para os quais A **não é invertível.**

$\det(A) = 0$

$$\det(A) = (x)(x+1) - (2 \times 3)$$

$$= (x^2 + 1x) - (6)$$

$$= x^2 + 1x - 6 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-6)}}{2} =$$

$$= \frac{-1 \pm \sqrt{1 - 4 \times 1 \times (-6)}}{2}$$

$$= \frac{-1 \pm \sqrt{1 - 4 \times (-6)}}{2}$$

$$= \frac{-1 \pm \sqrt{1 - (-24)}}{2}$$

$$= \frac{-1 \pm \sqrt{25}}{2}$$

$$x = \frac{-1 \pm 5}{2}$$

$$x_1 = \frac{-1+5}{2} = \frac{4}{2} = 2$$

$\vee$

$$x_2 = \frac{-1-5}{2} = \frac{-6}{2} = -3$$